FRONT / ESC1 — M1

SPI1 / BMI270 — FIRMWARE CONFIGURATION

Master, full duplex, 8-bit, MSB first; Mode 0: CPOL=0, CPHA=0.

Software NSS; SPI1 clock 72 MHz / 32 = SCK 2.25 MHz.

PA4: GPIO output push-pull, initial High; CS active Low.

Select SPI and initialize BMI270 per datasheet / Bosch SensorAPI.

Configuration notes only; firmware has NOT been created or flashed.

U1 INT1 / INT2 / SDX / SCX: NC; PA0 unused (poll SPI).

VERIFY U1.8 SDX / U1.9 SCX are auxiliary ASDx / ASCx on V1.3;

do not confuse these with Bosch chip primary SDx / SCx.

U7: logical module symbol; physical pin numbers / footprint UNVERIFIED.

M1
A2212/10T 1400KV

BT1
OVONIC LiPo 3S 11.1V

5200mAh / 90C
351P / 57.72Wh

U3
ESC 30A BLDC

U7
LM2596 module — 5.0V

LM2596 MODULE
SET OUT = 5.0 V
LOGICAL PIN IDs ONLY

PWR_FLAG

ESC4 — M4

M4
A2212/10T 1400KV

U6
ESC 30A BLDC

PWR_FLAG

U1
BMI270_DevKit_V1_3

BMI270
V1.3

SDX
SCX
INT2
VCC
GND
ADO/MISO
SDA/MOSI
SCL/SCLK
CS
INT1

USB UP

A1
STM32F103VET6_DevKit

STM32F103VET6 DEV KIT

LEFT 2x18 / RIGHT 2x18

BOTTOM 2x10 / SWD + BOOT

0 = OUTER / I = INNER

ESC2 — M2

U4
ESC 30A BLDC

M2
A2212/10T 1400KV

U2
SB_Nano_2G4RX_Physical

RX
TX
5V
GND

ELRS 2.4 GHz / UART1

U5
ESC 30A BLDC

M3
A2212/10T 1400KV

ESC3 — M3

CONTROL / UNUSED PINS

USART1: CRSF, 420000 baud, 8N1, non-inverted, full duplex;

no hardware flow control. PA9 TX -> RX; PA10 RX <- TX.

TIM4 default, no remap: B6 / B7 / B8 / B9 = ESC1 / 2 / 3 / 4.

PWM frequency, pulse range and calibration: match actual ESC.

400 Hz and DShot support are NOT assumed.

All four ESC pin 7 BEC/VCC: NC. No BEC or USB logic supply.

PA13/PA14 reserved for SWD; set existing BOOT0 jumper = 0.

Motor numbers do not define rotation or mixer.

Crossings connect only where a junction dot is shown.

Two PWR_FLAGs: ERC declarations only; no hardware, BOM or PCB.

POWER / HARDWARE VERIFICATION

Adjust AND measure LM2596 output = 5.0 V before connecting loads.

Verify DevKit accepts power at 5V and LM2596 has enough current.

Verify BMI270 V1.3 module supports a 3.3 V supply.

LiPo 3S: nominal 11.1 V; fully charged 12.6 V.

Never connect LiPo to STM32 5V, 3V3 or backup VBAT.

Never connect LM2596 5 V output to 3V3.

USB is NOT the supply; verify DevKit backfeed protection

before simultaneous USB connection.

Direct wires retained; two PWR_FLAG declarations for ERC only

Sheet: /
File: Drone_Direct_Wire.kicad_sch

Title: QUADROTOR — DIRECT MODULE WIRING

Size: A2	Date: 2026-11-02	**Rev: 2.1**
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